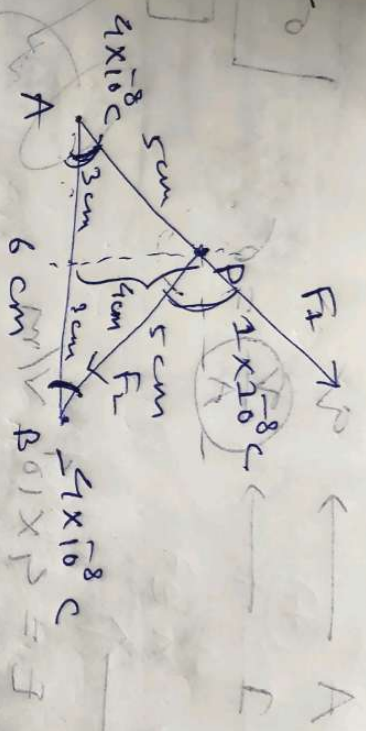


$\frac{u \pi^1}{A}$



$F_D = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} = F_{D1} + F_{D2}$

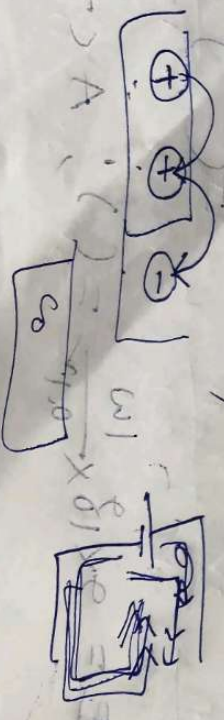
$F_L = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} = ?$ PB zero

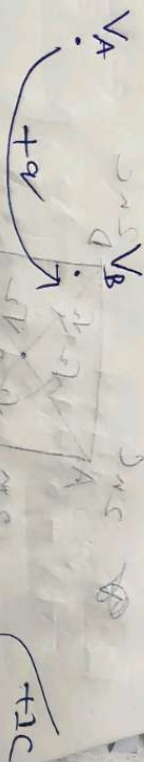
$\frac{F}{A} = \frac{1}{3}$

$A = \frac{1}{3} \left(\frac{1}{3} \right) = 0.53, 13^\circ$

off for

(T) ve $\rightarrow$ A high low





$$W = Vq = (V_A - V_B)q$$

$$\frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_A} + \frac{1}{r_B} \right) + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_B} + \frac{1}{r_C} \right)$$

$$+q \rightarrow W = Vq$$

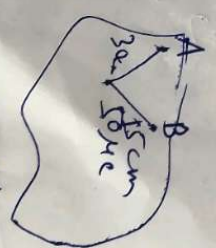


$$V = \frac{W}{q} = \frac{1}{q}$$

$$\frac{1}{q} \rightarrow V$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

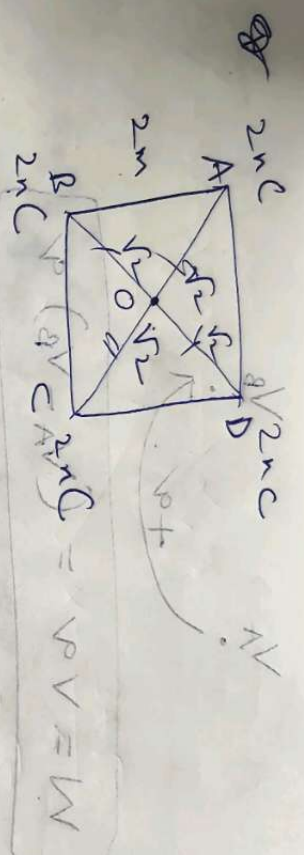
30 cm $\rightarrow A$



$$V_A = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{50 \times 10^{-2} \text{ m}}$$

$$\left(\frac{q}{4\pi\epsilon_0} + \frac{q}{4\pi\epsilon_0} \right) + \frac{Q}{4\pi\epsilon_0} = W = (V_A - V_B)q$$

$$q + q + Q = 0$$

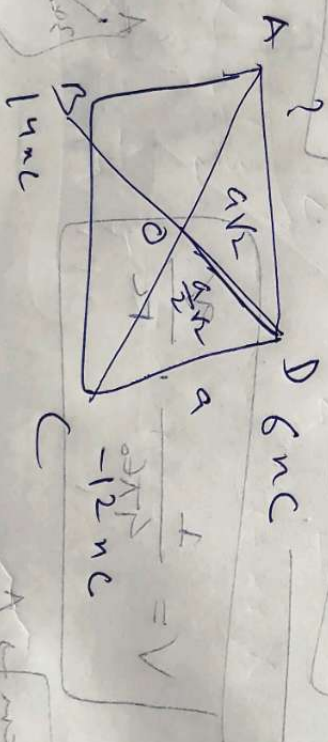


$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_A}{r_{OA}} + \frac{q_B}{r_{OB}} + \frac{q_C}{r_{OC}} + \frac{q_D}{r_{OD}} \right)$$

$$= 9 \times 10^9 \times \frac{2 \times 10^{-9}}{\sqrt{2}} (4 \times 2 \times 10^{-3})$$

$$V = 50.92 \text{ V}$$

$$V = 50.92 \text{ V}$$



$$V = 10V$$

$$0 = \frac{1}{4\pi\epsilon_0} \left(\frac{q_A}{r_{OA}} + \frac{q_B}{r_{OB}} + \frac{q_C}{r_{OC}} + \frac{q_D}{r_{OD}} \right)$$

$$0 = q_A + q_B + q_C + q_D$$

$$a^0 \quad a^1 \quad a^2 \quad a^3$$

$$x = \frac{a+ib}{a-ib}$$

$$\frac{2x+5}{3+12} = \frac{1}{3}$$

$$a_1 x + b_1 x + c_1 = 0 \rightarrow a \textcircled{a}$$

$$a_2 x + b_2 x + c_2 = 0 \rightarrow a \textcircled{b}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$ax + bx + c = 0$$

$$(x-\alpha) = (x-\beta)$$

$$(x-\alpha)(x-\beta) = 0$$

$$x - \alpha x - \beta x + \alpha\beta = 0$$

$$x^2 - (\alpha+\beta)x + \alpha\beta = 0$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

$$x + \beta = -\frac{b}{a}, \quad x\beta = \frac{c}{a}$$

$$4x^2 - 6x + 1 = 0 \rightarrow x, \beta$$

$$x + \frac{1}{\beta}$$

$$\beta + \frac{1}{x}$$

$$x = \frac{3 + \sqrt{5}}{4}$$

$$\beta = \frac{3 - \sqrt{5}}{4}$$

$$(x - \gamma)(x - \theta) = 0$$

$$4x^2 - 6x + 1 = 0, \boxed{x, \beta}$$

$$x + \beta = -\frac{-b}{a} = \frac{6}{4} = \frac{3}{2}$$

$$x + \beta = \frac{3}{2}$$

$$x\beta = \frac{1}{4}$$

$$x + \frac{1}{\beta} + \beta + \frac{1}{x} = (x + \frac{\beta}{x}) + (\frac{1}{x} + \frac{1}{\beta})$$

$$= (x + \beta) + (\frac{1}{x} + \frac{1}{\beta})$$

$$= \frac{3}{2} + \frac{1}{\frac{1}{4}} = \frac{3}{2} + 4 = \frac{11}{2}$$

$$= \frac{15}{2}$$

$$0 = 6 + x + x + x + x + x$$

$$\left(x + \frac{1}{x}\right) \left(x + \frac{1}{x}\right) = x^2 + 1 + 1 + \frac{1}{x^2}$$

$$x^2 + \frac{1}{x^2} + 2 + \frac{4}{x}$$

$$\frac{2}{x} = \frac{4 \cdot 25}{4} = 25$$

$$\frac{4}{x} = 25$$

$$x - \frac{15}{2}x + \frac{25}{4} = 0$$

$$4x^2 - 30x + 25 = 0$$

AA

$$x - (35x) + (25x) = 0$$

$$x + x + x + x + x + x$$

$$x + x + x + x + x + x$$

$$\frac{21}{2} = 10.5$$

$$\frac{21}{2} = 10.5$$

2
-
w
-
2

(11)

$$\frac{c}{r}$$

$$\frac{1}{\omega}$$

0
11
2/5

$$0 = \alpha \beta \gamma - 1$$

$$E_{\text{kin}} = 0$$

230505
Cory Johnson

(contd)

$$\sqrt{-4(a-b)} \rightarrow 0$$

$$a=b$$

$$\left[\frac{3x - a}{x} \right]$$

$$(a-b)x^2 + 2(a+b)x + a-b = 0$$

①

$$\text{①} \rightarrow \frac{1}{x^2}$$

first term

$$\text{①} \rightarrow \text{2nd term}$$

$$= 4(a+b) - 4(a-b)$$

$$= 4 \{ (a+b) - (a-b) \}$$

$$= 4 \{ 4ab \}$$

$$= 16ab$$

$$= (4ab)^2$$

2nd term

No. 13

$$ax^2 + bx + c = 0 \quad \text{--- (1)}$$

4x, 5x

माना x, p

$$\therefore \frac{x}{p} = \frac{5}{5}$$

$$\rightarrow x = \frac{5}{5} p$$

$$x + p = -\frac{5}{5}$$

$$p = -\frac{5}{5} - x$$

$$\Rightarrow \frac{5}{5} p + p = -\frac{5}{5}$$

$$p \left(1 + \frac{5}{5} \right) = -\frac{5}{5}$$

$$p \cdot \frac{2}{5} = -\frac{5}{5}$$

$$p = -\frac{5 \cdot 5}{2 \cdot 5}$$

$$x = \frac{5}{5} p$$

$$\frac{5}{5} p = \frac{5}{5} p$$

$$p = \frac{5}{5}$$

$$p = \sqrt{\frac{5}{5}}$$

$$\sqrt{\frac{5}{5}} = -\frac{5}{5}$$

$$\frac{5}{5} = \frac{25}{5}$$

$$2067 = 5100$$

15

$$ax + bx + b = 0$$

$$\frac{a}{x} = \frac{b}{x}$$

$$\frac{a}{x} = \frac{b}{x}$$

$$L.H.S. = \frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$L.H.S. = \frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\frac{a}{x} + \frac{b}{x}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$ax^2 + bx + c = 0 \Rightarrow x + x + x$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$x + y + z = 1$$

$$x \cdot y \cdot z = \frac{c}{a}$$

$$x - y = (x + y) \cdot x$$

$$\frac{b}{a} = x \cdot y$$

$$x = \frac{b}{a(x+1)}$$

$$x \cdot \frac{b}{a(x+1)} = \frac{c}{a}$$

$$\frac{b}{a(x+1)} = \frac{c}{a}$$

$$\frac{1}{x+1} = \frac{c}{b}$$

$$= \sqrt{\frac{b}{c}} + \sqrt{\frac{b}{c}} + \sqrt{\frac{b}{c}}$$

$$+ \sqrt{\frac{b}{c}}$$

$$\frac{1}{x+1} = \frac{c}{b}$$